

Oracle CDC PoC - Project Offer

Project: Oracle Change Data Capture - Proof of Concept
Date: 2026-02-10
Version: 3.0 - Object Store Only, Simplified Structure
Prepared by: P3 Team (Matthias Max)
Client: Nagel IT
Decision Maker: Christian Lang

1. Executive Summary

1.1 Background

NewDispo requires Oracle CDC integration to be deployed independently from Project G. Striim is currently deployed on 5 Oracle branches (historically on all branches) for CDC purposes. Christian Lang has requested a comprehensive Proof of Concept to evaluate two CDC approaches:

1. **Option A: Striim → GCP Object Store** (leveraging existing Striim infrastructure)
2. **Option B: Oracle → GCP Datastream → Object Store** (GCP-native solution)

1.2 Project Objectives

- Validate end-to-end CDC functionality for both options on Nagel IT infrastructure
- Determine implementation effort, costs, and operational requirements for each option
- Define clear rollout steps for global deployment across all Oracle branches
- Provide decision-making basis for strategic CDC solution selection via ADR

1.3 Success Criteria

Criterion	Target
CDC Solution Identified	Suitable option selected based on performance, cost, and operational fit

Criterion	Target
Change Capture Validated	Table changes (INSERT, UPDATE, DELETE) successfully captured and written to Object Store
Setup Documented	Complete documentation for future rollout on client infrastructure
SE2 Compatibility	Validation of Oracle Standard Edition 2 support for selected option
Cost Model	Clear production cost projection for both options

2. PoC Scope & Constraints

2.1 Constraints

Constraint	Rationale
TMS Branch Databases Only	PoC focuses on branch databases (Standard Edition 2) which represent the majority of the deployment target (~30 branches)
PoC on Nagel IT Infrastructure	End-to-end validation on actual Nagel Oracle and GCP environment; requires separate/dedicated environment to not block testing on TEST
Nagel IT Provides Infrastructure	Nagel IT responsible for providing Oracle databases meeting load criteria and GCP connectivity
Two Separate Branch Oracle Databases	ONE Striim-connected branch DB and ONE non-Striim branch DB to avoid dual-CDC conflicts
Fully Fledged Database with Single Table Focus	Use actual production-like branch database, but constrain PoC testing to "sendung" table only
PoC Ends at Object Store	Full-stack chain validates: Oracle → CDC Tool → Object Store (GCS)
No GoLive Until PROD-Readiness	Technical evaluation only; production deployment is separate phase

Constraint	Rationale
Network Already Established	Leverage existing VPN/Interconnect to GCP (provided by Nagel IT)

2.2 In Scope

Technical Scope:

- Test Oracle databases provided by Nagel IT (archive log mode enabled, LogMiner configured)
- Configuration and testing of Striim → Object Store integration
- Configuration and testing of Oracle → Datastream → Object Store integration
- Performance and latency measurements for both options
- Oracle Standard Edition 2 compatibility validation (via documentation/Google support)
- Failure scenario testing (recovery, redo log gap handling)
- Cost analysis and projection for production deployment

Deliverables:

- Working PoC for both options
- ADR (Architecture Decision Record) with comparative evaluation
- Implementation guides for both options
- Production rollout plan (effort estimates and steps) - **TBD with Martin**
- Cost comparison and operational complexity assessment

2.3 Out of Scope

- **Central HQ Oracle database** (Enterprise Edition) - PoC focuses on branch databases (Standard Edition 2) only
- Production deployment on Nagel infrastructure (covered in rollout plan)
- Integration with actual NewDispo consumers (simulated in PoC)
- Cross-Dock/CALSuite integration
- Full decommissioning of non-selected option (plan provided, execution separate)

2.4 Test Setup

Oracle Databases (Nagel Infrastructure):

IMPORTANT: Dedicated/Separate Environment Required

- PoC requires dedicated Oracle database environment (not TEST environment)
- Ensures PoC activities don't block or interfere with ongoing TEST activities
- Nagel IT to provide selected branch database instances for PoC

Two separate **branch** Oracle databases will be used to avoid dual-CDC conflicts:

1. **For Striim PoC:**

- Database: ONE of the 5 existing Striim-connected **TMS branch** Oracle databases
- Edition: Standard Edition 2 (branch database)
- Purpose: Validate Striim extension to Object Store (GCS)
- Configuration: Already configured with Striim CDC
- Test Table: "sendung" table

2. **For Datastream PoC:**

- Database: ONE **TMS branch** Oracle database that does NOT have Striim configured
- Edition: Standard Edition 2 (branch database)
- Purpose: Validate Datastream CDC independently on SE2
- Configuration: Archivelog mode, LogMiner enabled, supplemental logging, necessary grants, and all further CDC-required configurations (to be validated with Nagel IT DB experts - support may be required)
- Test Table: "sendung" table

Rationale for Branch Database Focus:

- Branch databases use **Standard Edition 2** with LogMiner limitations that need validation
- Branch databases represent the majority of the deployment target (~30 branches)
- SE2 compatibility is critical for production rollout success

GCP Environment (Nagel GCP Project):

- Cloud Storage buckets (Object Store) for both PoCs
- Datastream service configuration (for Option B)
- Monitoring and logging setup
- Existing network connectivity (VPN/Interconnect) leveraged

2.5 Load & Volume Requirements

Understanding the expected load and volume is critical for validating PoC performance and projecting production costs.

Key Questions to Address During Kickoff:

Question	Rationale	Status
How many shipments (Sendungen) per day?	Determines CDC event volume and throughput requirements	TBD - Nagel IT
Peak vs. low times during the day?	Identifies if solution must handle burst traffic or steady load	TBD - Nagel IT
Average Sendung record size?	Impacts Datastream per-GiB cost projections	TBD - Nagel IT
INSERT/UPDATE/DELETE ratio?	Different operations may have different CDC characteristics	TBD - Nagel IT
Does existing Oracle-Striim handle high-load?	Validates if current Striim deployment can scale for additional targets	TBD - Matt Wilkinson
Current Striim throughput metrics?	Baseline for comparison; validates if extension is viable	TBD - Matt Wilkinson

Initial Assumptions (To Be Validated):

- **Volume Estimate:** Assuming 10,000 - 100,000 shipments/day across all branches (TBD)
- **CDC Event Size:** Assuming average 2-5 KB per Sendung CDC event (TBD)
- **Peak Load:** Assuming 2-3x average during peak hours (TBD)
- **Daily Data Volume:** Estimated 20-500 MB/day CDC data (TBD)

Impact on PoC:

1. **Performance Testing:** PoC will simulate expected load to validate latency under realistic conditions
2. **Cost Projections:** Accurate volume data enables precise Datastream cost estimates (per-GiB pricing)
3. **Striim Capacity:** Validates if existing Striim can handle additional Object Store target without performance degradation
4. **Sizing Recommendations:** Informs production deployment sizing (GCP resources, network bandwidth)

Note: These questions will be addressed during the kickoff meeting. If accurate production metrics are unavailable, PoC will use conservative estimates and provide sensitivity analysis for cost projections.

3. Technical Approach

3.1 Option A: Striim → GCP Object Store

Architecture (Client Infrastructure):

Oracle DB (Nagel On-Prem) → Existing Striim → Object Store (GCS)

Key Activities:

1. Assess existing Striim deployment and configuration
2. Configure Object Store (GCS) writers in Striim
3. Configure CDC for "sendung" table
4. Run INSERT/UPDATE/DELETE operations and validate capture
5. Measure latency and throughput
6. Test failure scenarios (network interruption, redo log gap)
7. Document operational procedures

Key Validations:

- Striim can capture Oracle changes and publish to Object Store
- Latency meets requirements (target: < 5 seconds 95th percentile)
- Operational complexity is manageable
- Cost model for production (obtain Striim licensing costs from Nagel IT)

3.2 Option B: Oracle → GCP Datastream → Object Store

Architecture (Client Infrastructure):

Oracle DB (Nagel On-Prem) → Datastream → Object Store (GCS)

Key Activities:

1. Validate Datastream Oracle Standard Edition 2 support (documentation + Google support)
2. Configure Datastream connection profile for Oracle
3. Set up Datastream stream for "sendung" table
4. Configure GCS bucket as target

- 5. Run INSERT/UPDATE/DELETE operations and validate capture
- 6. Measure latency and throughput (including GCS write latency)
- 7. Test failure scenarios
- 8. Document operational procedures

Key Validations:

- Datastream supports Oracle SE2 with required LogMiner features
- End-to-end latency is acceptable (note: GCS intermediate step adds latency)
- Cost model for production (per-GiB pricing)
- Operational simplicity (fully managed service)

3.3 Comparative Evaluation Criteria

Criterion	Weight	Notes
Technical Feasibility	High	SE2 compatibility, performance, reliability
Cost (PoC + Production)	High	Licensing costs (Striim) vs. volume-based costs (Datastream)
Operational Complexity	Medium	Setup, maintenance, monitoring, troubleshooting
Strategic Alignment	Medium	Existing Striim investment vs. GCP-native strategy
Vendor Lock-in Risk	Low	Dependency on third-party (Striim) vs. GCP
Time to Production	Medium	Effort to deploy on Nagel infrastructure

4. Key Technical Challenges & Mitigations

4.1 Oracle Standard Edition 2 Limitations

Challenge: Oracle SE2 does not support all LogMiner features:

- No Continuous LogMiner (streaming API)
- Limited concurrent LogMiner sessions
- Supplemental Logging restrictions

Impact: Datastream (and potentially Striim) may have degraded functionality or incompatibility on SE2.

Mitigation:

- **Phase 1:** Validate Datastream SE2 support via Google documentation and support ticket
- **PoC:** Test both options with clean Oracle SE2 install
- **Fallback:** If Datastream incompatible with SE2, Striim becomes primary option (already proven on Nagel SE2 branches)

4.2 Redo Log Retention (~1 hour on some branches)

Challenge: Nagel branches (e.g., D33) have ~1 hour redo log retention due to space constraints.

Impact: Network outage > 1 hour results in transaction sequence loss, requiring full initial load.

Mitigation:

- Design CDC solution with recovery/resync capability
- Document operational procedures for full reload scenarios
- **PoC:** Test redo log gap scenario and document recovery steps

4.3 Dual-CDC Risk

Challenge: Running both Striim and Datastream simultaneously risks data conflicts and infrastructure complexity.

Mitigation:

- **Clear Decision via ADR:** Select one option based on PoC results
- **Decommissioning Plan:** Before production rollout of new option, coordinate Striim retirement with Nagel IT
- **Database Separation:** Using separate Oracle databases for Striim and Datastream PoCs avoids dual-CDC conflicts

4.4 Cost Uncertainty

Challenge: Striim licensing costs for extending to Object Store are unknown. New Striim deployment costs \$9,600+/month.

Impact: If Striim extension costs are prohibitive, Datastream may be more cost-effective.

Mitigation:

- **Immediate Action:** Obtain Striim cost data from Nagel IT (Matt Wilkinson) before/during PoC
- **Cost Modeling:** Project production costs for both options based on:
 - Striim: licensing + infrastructure
 - Datastream: per-GiB pricing (~\$2/GiB) based on expected data volume

5. Deliverables

5.1 Technical Documentation

1. PoC Setup Guide

- Oracle database installation and configuration
- Striim deployment and configuration guide
- Datastream setup and configuration guide
- GCP infrastructure (GCS, networking) setup

2. Performance Test Results

- Latency measurements (95th percentile, average)
- Throughput measurements (events/second)
- Resource utilization (CPU, memory, network)
- Failure scenario test results (recovery time, data loss)

3. Operational Runbooks

- Day-to-day operations procedures
- Monitoring and alerting setup
- Troubleshooting guide (common issues and resolution)
- Failure recovery procedures (redo log gap, network outage)

5.2 Architecture Decision Record (ADR)

Comprehensive ADR document including:

1. Context & Requirements

- Business objectives and constraints
- Technical environment (Oracle versions, editions, infrastructure)

2. Options Evaluated

- Option A: Striim → Object Store (detailed analysis)
- Option B: Datastream → Object Store (detailed analysis)

- Options considered but rejected (e.g., Debezium, GoldenGate)

3. **Comparative Evaluation**

- Performance comparison (latency, throughput)
- Cost analysis (PoC + projected production costs)
- Operational complexity assessment
- Risk and dependency analysis
- Strategic alignment considerations

4. **Recommendation**

- Recommended option with clear justification
- Trade-offs and considerations
- Implementation roadmap

5. **Consequences**

- Impact of decision on architecture
- Required follow-up actions
- Risks and mitigation strategies

5.3 Production Rollout Plan

Note: Detailed production rollout plan (effort estimates, timeline, and steps) is **TBD with Martin**. The following outlines the expected structure and key areas to be covered:

1. **Deployment Strategy**

- Phase-by-phase rollout approach (pilot branches → full rollout)
- Branch prioritization (based on criticality, Oracle version/edition)
- Rollback procedures and contingency plans

2. **Effort Estimates**

- Per-branch deployment effort (setup, testing, cutover)
- Total effort for all Oracle branches (~30 branches estimated)
- Timeline projections (parallelization opportunities)

3. **Prerequisites & Dependencies**

- Network connectivity requirements
- Oracle database grants and configuration
- GCP infrastructure setup
- Stakeholder approvals and coordination

4. **Risk Assessment**

- Technical risks (SE2 compatibility, redo log gaps)
- Operational risks (dual-CDC conflicts, downtime)
- Mitigation strategies and contingency plans

5. Decommissioning Plan (if applicable)

- Steps to retire non-selected CDC option
- Data validation during transition
- Rollback procedures if issues arise

6. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation
Datastream incompatible with Oracle SE2	Medium	High	Early validation with Google; Striim as proven fallback
Striim extension costs prohibitive	Medium	High	Obtain cost data early in PoC; shift focus to Datastream if needed
Short redo log retention causes production issues	High	Medium	Design for resync capability; operational procedures documented
Dual-CDC conflicts on Nagel infrastructure	Low	High	Clear ADR decision; decommissioning plan before production rollout
PoC blocks TEST environment or causes disruption	Low	Medium	Use dedicated/separate environment; coordinate with Nagel IT on environment allocation
Latency exceeds NewDispo requirements	Low	Medium	Performance testing in PoC; tuning and optimization
Access delays to Nagel Striim cost data	Medium	Low	Immediate request to Matt Wilkinson; proceed with PoC in parallel

7. Success Metrics & Acceptance Criteria

7.1 PoC Success Metrics

Metric	Target	Measurement
CDC Event Capture Accuracy	100%	Validate all INSERT/UPDATE/DELETE captured
End-to-End Latency	< 5 seconds (95th percentile)	Measure from Oracle commit to Object Store write
Initial Load Performance	< 1 hour for test dataset	Validate backfill mechanism
Recovery Time (Redo Log Gap)	< 30 minutes	Test resync after simulated outage
SE2 Compatibility	Full functionality	Validate on Oracle SE2 instance
Operational Complexity	Manageable by Ops team	Documented procedures, clear runbooks
Cost Predictability	±20% accuracy	Production cost projections validated

7.2 ADR Acceptance Criteria

- ☐ Both options evaluated with objective criteria
- ☐ Performance data collected and analyzed
- ☐ Cost comparison completed (PoC + 12-month production projection)
- ☐ Operational complexity assessed for both options
- ☐ Clear recommendation with justification
- ☐ Risk assessment and mitigation strategies documented
- ☐ Stakeholder review and approval obtained

7.3 Rollout Plan Acceptance Criteria

Note: Detailed acceptance criteria are **TBD with Martin**. Preliminary criteria include:

- ☐ Phase-by-phase deployment strategy framework defined
- ☐ Preliminary per-branch effort estimates provided

- ☐ Indicative rollout timeline projected
- ☐ Key prerequisites and dependencies identified
- ☐ Risk mitigation and rollback procedures documented
- ☐ Decommissioning plan for non-selected option (if applicable)

8. Assumptions & Dependencies

8.1 Assumptions

- **Client Infrastructure:** Nagel IT provides dedicated/separate infrastructure meeting load criteria and GCP connectivity
- **Infrastructure Responsibility:** Nagel IT responsible for:
 - Providing dedicated/separate PoC environment (not TEST environment)
 - Providing two Oracle branch databases (one Striim-connected, one non-Striim)
 - Ensuring databases meet expected load characteristics (Sendungen volume)
 - Maintaining network connectivity to GCP (VPN/Interconnect)
 - Oracle configuration (archive log mode, LogMiner enabled, supplemental logging, necessary grants, and all CDC-required configurations; DB expert support from Nagel IT available)
- **P3 Team Access:** P3 has access to Nagel Oracle databases, Striim, and GCP project
- **Separate PoC Environment:** Dedicated/separate Oracle database environment provided by Nagel IT that won't block TEST environment or disrupt production
- **Striim Access:** Existing Striim deployment accessible for configuration changes
- **Internal Support Guaranteed:** Support requests from Nagel IT, Platform, and BIT teams are guaranteed and prioritized by Christian Lang throughout PoC duration
- **Vendor Support Availability:**
 - Striim vendor support accessible if needed (via Nagel IT's support contract)
 - Google Cloud support available for Datastream questions (reasonable response times: 24-48h)
 - Critical issues can be escalated
- **Data Volume Estimates:** Nagel IT can provide expected CDC data volume for cost projections
- **Fully Fledged Database:** PoC uses actual production-like branch database with full TMS-Schema; testing constrained to "sendung" table only

8.2 Dependencies

Dependency	Owner	Required By	Status
Infrastructure Provision: Nagel IT provides dedicated/separate PoC environment meeting load criteria (not TEST)	Nagel IT	Week 1	Pending
GCP Connectivity: Nagel IT ensures stable VPN/Interconnect to GCP	Nagel IT / DevOps	Week 1	Pending
Oracle DB #1: Access to ONE Striim-connected database	Nagel IT / DBA Team	Week 1	Pending
Oracle DB #2: Access to ONE non-Striim Oracle database (preferably SE2)	Nagel IT / DBA Team	Week 1	Pending
Nagel GCP project access for P3 team	Nagel IT / DevOps	Week 1	Pending
Striim access and configuration rights	Nagel IT (Matt Wilkinson)	Week 1	Pending
Striim cost data (current + Object Store extension)	Nagel IT (Matt Wilkinson)	Week 1-2	Pending
Striim performance metrics (throughput, current load)	Nagel IT (Matt Wilkinson)	Week 1-2	Pending
Striim vendor support access (if needed for troubleshooting)	Nagel IT (Matt Wilkinson)	Week 2-3	Pending
Datastream SE2 compatibility confirmation	Google Cloud Support	Week 1-2	Pending
Google Cloud expert support access for Datastream setup	P3 / Nagel IT	Week 2-3	Pending
Expected CDC data volume estimates (Sendungen/day, sizes)	Nagel IT	Week 1-2	Pending
Stakeholder availability for meetings	Nagel IT (Christian, Matt, Robert)	Throughout PoC	Pending

Dependency	Owner	Required By	Status
Internal support from Nagel IT, Platform, BIT teams guaranteed and prioritized	Christian Lang	Throughout PoC	Pending

9. Appendix

9.1 Key Stakeholders

Name	Organization	Role	Responsibility
Christian Lang	Nagel IT	Decision Maker	Strategic direction, final ADR approval
Matt Wilkinson	Nagel IT	Striim Lead	Striim cost data, technical guidance
Robert Zanter	Nagel IT	DBA Lead	Oracle technical validation, LogMiner insights
Matthias Max	P3	Architect	PoC architecture, ADR authoring, evaluation
[TBD]	P3	DB-Dev Lead	Oracle setup, CDC configuration, testing
[TBD]	P3	DevOps Lead	GCP infrastructure, networking, integration
[TBD]	P3	PM	Project coordination, reporting, stakeholder management

9.2 Reference Documents

- [Original Request \(2025-12-01\)](#)
- [Consolidated Requirements](#)
- [Architecture Status Update \(2026-01-16\)](#)
- [Architecture Evaluation](#)
- [P3 Standard PoC Template](#)

9.3 Glossary

Term	Definition
ADR	Architecture Decision Record - document capturing options, evaluation, and decision
CDC	Change Data Capture - technology to track and capture database changes
LogMiner	Oracle feature to read redo logs for change tracking
SE2	Oracle Standard Edition 2 (vs. Enterprise Edition) - has LogMiner limitations
Redo Log	Oracle transaction log used for recovery and CDC
Datastream	Google Cloud service for database replication and CDC
Striim	Third-party real-time data integration platform (currently deployed at Nagel)
Object Store	Cloud storage (e.g., GCS) for persisting CDC events
GCS	Google Cloud Storage
End-to-End PoC	PoC validates full CDC flow on actual client infrastructure (not isolated sandbox)

9.4 Oracle SE2 LogMiner Limitations

Oracle Standard Edition 2 has the following LogMiner restrictions that may impact CDC tools:

Feature	Enterprise Edition	Standard Edition 2
Continuous LogMiner	Supported	Not Supported
Concurrent LogMiner Sessions	Unlimited	Limited (typically 1-2)
Supplemental Logging (All Columns)	Supported	Limited in some configurations
Online Catalog	Supported	Supported
Dictionary Extraction	Supported	Supported

Impact: CDC tools that rely on Continuous LogMiner API (streaming mode) may require polling-based approach on SE2, potentially affecting latency and resource usage.

Validation Required: Confirm Datastream's SE2 support via Google Cloud documentation and support ticket during PoC preparation phase.

10. Approval

Role	Name	Signature	Date
Client Approver	Christian Lang		
P3 Project Lead	Martin Dittmann		

Document Version History:

Version	Date	Author	Changes
1.0	2026-02-10	Matthias Max	Initial draft
2.0	2026-02-10	Matthias Max	Aligned with P3 standard PoC template, adjusted scope and effort
3.0	2026-02-16	Matthias Max	Changed to Object Store only (removed Pub/Sub), removed sections: Project Activities & Effort, Cost Analysis Document, Cost Breakdown, Next Steps